

Cybersecurity Training Diary

Day 26: Cloud Security & Hardening

1. Topics Covered

✓ Cloud Shared Responsibility Model

```
graph LR
    A[Customer] -->|Secure| B[Data/Apps]
    C[Provider] -->|Secure| D[Hardware/Global Infrastructure]
```

✓ Top Cloud Threats (2024)

Risk	Example	Mitigation
Misconfigurations	Exposed S3 buckets	CSPM tools (Prisma Cloud)
IAM Overprivilege	Overly permissive roles	Least privilege, PIM
API Vulnerabilities	Unauthenticated endpoints	WAF, API gateways

✓ Multi-Cloud Security Tools

- **CNAPP:** Wiz, Orca
- **IaC Scanning:** Checkov, Terrascan
- **Secrets Detection:** GitLeaks, TruffleHog

2. Key Takeaways

- ✓ **90% of cloud breaches** stem from misconfigurations (Gartner)
- ✓ **Identity is the new perimeter** - focus on IAM policies
- ✓ **Shift-left for cloud:** Scan infrastructure-as-code pre-deployment

3. Practical Exercises

♦ Exercise 1: Find Exposed S3 Buckets

```
aws s3 ls --no-sign-request # Check for public buckets
s3scanner scan --buckets-file buckets.txt
```

♦ Exercise 2: Harden AWS Root Account

```
# Enable MFA and restrict root
aws iam create-virtual-mfa-device --virtual-mfa-device-name MyMFADevice
aws iam enable-mfa-device --user-name root --serial-number arn:aws:iam::123456789012:mfa/root --auth-
```

4. Questions for Reflection

- ? Why do cloud providers default to insecure settings?
- ? How does cloud security differ from traditional datacenter security?